

Filter banks localization

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1 Standard EnVar formulation

For an ensemble N backgrounds $\{\mathbf{x}^1, \dots, \mathbf{x}^N\}$, with $\mathbf{x}^k \in \mathbb{R}^n$, the standard EnVar formulation is given by:

$$\mathbf{B} = \mathbf{X}\mathbf{X}^T \circ \mathbf{L} \quad (1)$$

where:

- $\mathbf{B} \in \mathbb{R}^{n \times n}$ is the resulting background error covariance.
- $\mathbf{X} \in \mathbb{R}^{n \times N}$ contains the normalized ensemble perturbations:

$$X_{ij} = \frac{1}{\sqrt{N-1}} \left(x_i^j - \langle x_i \rangle \right) \quad (2)$$

with $\langle x_i \rangle$ denoting the ensemble mean:

$$\langle x_i \rangle = \frac{1}{N} \sum_{j=1}^N x_i^j \quad (3)$$

- $\mathbf{L} \in \mathbb{R}^{n \times n}$ is the localization matrix.
- $\circ$ denotes a Schur product between matrices (element-by-element).

The square-root expression of (1), transforming the control vector $\mathbf{v} \in \mathbb{R}^m$ into a control increment $\delta\mathbf{x} \in \mathbb{R}^n$, is given by:

$$\delta\mathbf{x} = \sum_{j=1}^N \mathbf{X}_{:j} \circ \left(\mathbf{U}\mathbf{v}_j \right) \quad (4)$$

where:

- $\mathbf{X}_{:j} \in \mathbb{R}^n$ is the j^{th} column of $\mathbf{X}$
- $\mathbf{U} \in \mathbb{R}^{n \times m}$ is a square-root of the localization $\mathbf{L}$ such that $\mathbf{U}\mathbf{U}^T = \mathbf{L}$.
- $\mathbf{v}_j \in \mathbb{R}^{m/N}$ is the j^{th} subvector of $\mathbf{v}$.

2 Generalized EnVar formulation

The standard EnVar formulation can be generalized by introducing two linear transforms $\mathbf{F} \in \mathbb{R}^{\hat{n} \times n}$ and $\mathbf{F}^\dagger \in \mathbb{R}^{n \times \hat{n}}$:

$$\hat{\mathbf{B}} = \mathbf{F}^\dagger \left(\mathbf{F} (\mathbf{X}\mathbf{X}^\top) \mathbf{F}^\top \circ \hat{\mathbf{L}} \right) (\mathbf{F}^\dagger)^\top \quad (5)$$

$$\delta \mathbf{x} = \mathbf{F}^\dagger \sum_{j=1}^N \left(\mathbf{F} \mathbf{X}_{:j} \right) \circ \left(\hat{\mathbf{U}} \hat{\mathbf{v}}_j \right) \quad (6)$$

where:

- $\hat{\mathbf{L}} \in \mathbb{R}^{\hat{n} \times \hat{n}}$ is the localization matrix in the transformed space.
- $\hat{\mathbf{U}} \in \mathbb{R}^{n \times \hat{m}}$ is a square-root of the localization $\hat{\mathbf{L}}$ such that $\hat{\mathbf{U}} \hat{\mathbf{U}}^\top = \hat{\mathbf{L}}$.
- $\hat{\mathbf{v}}_j \in \mathbb{R}^{\hat{m}/N}$ is the j^{th} subvector of $\hat{\mathbf{v}} \in \mathbb{R}^{\hat{m}}$.

Let's analyse the asymptotic values if the ensemble size grows to infinity:

- The localization matrices $\mathbf{L}$ and $\hat{\mathbf{L}}$ should converge to $\mathbf{1}_n$ and $\mathbf{1}_{\hat{n}}$, respectively, where $\mathbf{1}_n \in \mathbb{R}^{n \times n}$ is a matrix filled with 1.
- As a consequence:

$$\lim_{N \rightarrow \infty} \mathbf{B} = \mathbf{X}\mathbf{X}^\top \quad (7)$$

$$\lim_{N \rightarrow \infty} \hat{\mathbf{B}} = (\mathbf{F}^\dagger \mathbf{F} \mathbf{X}) (\mathbf{F}^\dagger \mathbf{F} \mathbf{X})^\top \quad (8)$$

- To ensure consistency, linear transforms $\mathbf{F}$ and $\mathbf{F}^\dagger$ should not have an impact on the asymptotic background error covariance:

$$\lim_{N \rightarrow \infty} \mathbf{B} = \lim_{N \rightarrow \infty} \hat{\mathbf{B}} \quad (9)$$

so

$$\mathbf{F}^\dagger \mathbf{F} \mathbf{X} = \mathbf{X} \quad (10)$$

Thus, $\mathbf{F}^\dagger$ should be a left-inverse of $\mathbf{F}$ in the ensemble perturbations space.

The content of $\mathbf{F}$ can be very diverse:

- A change of variables: for instance towards a better-suited humidity variable.
- A inverse balance operator as in Clayton *et al.* (2012), where the localization is applied on unbalanced variables and the balance operator is finally applied to get the increment.
- A change of resolution as in Kleist and Ide (2014), where the ensemble resolution is lower than the final increment one. In this case, $\mathbf{F}^\dagger$ is an interpolation from low to high resolution, and the low-resolution perturbations $\mathbf{X}_{\text{LR}}$ can be thought as the application of a coarsening operator $\mathbf{F}$ on virtual high-resolution perturbations $\mathbf{X}_{\text{HR}}$ representing the same field:

$$\mathbf{F}^\dagger \mathbf{X}_{\text{LR}} = \mathbf{F}^\dagger \mathbf{F} \mathbf{X}_{\text{HR}} = \mathbf{X}_{\text{HR}} \quad (11)$$

3 Filter banks localization

In the filter banks localization (FBL) framework, we define two series of $P - 1$ linear transforms:

- The low-pass filters $\{\mathbf{G}_1, \dots, \mathbf{G}_{P-1}\}$, where $\mathbf{G}_p \in \mathbb{R}^{\hat{n}_p \times n}$.
- The interpolators $\{\mathbf{S}_1, \dots, \mathbf{S}_{P-1}\}$, where $\mathbf{S}_p \in \mathbb{R}^{n \times \hat{n}_p}$.

The components of $\mathbf{F}$ are built recursively:

- The low-pass filter $\mathbf{G}_1$ is applied to the input $\mathbf{x} \in \mathbb{R}^n$ to produce $\hat{\mathbf{x}}_1 \in \mathbb{R}^{\hat{n}_1}$:

$$\hat{\mathbf{x}}_1 = \mathbf{G}_1 \mathbf{x} \quad (12)$$

- The residual is computed from $\hat{\mathbf{x}}_1$, interpolated at full resolution:

$$\bar{\mathbf{x}}_1 = \mathbf{x} - \mathbf{S}_1 \hat{\mathbf{x}}_1 \quad (13)$$

- The low-pass filter $\mathbf{G}_2$ is applied to $\bar{\mathbf{x}}_1$ to produce $\hat{\mathbf{x}}_2 \in \mathbb{R}^{\hat{n}_2}$:

$$\hat{\mathbf{x}}_2 = \mathbf{G}_2 \bar{\mathbf{x}}_1 \quad (14)$$

- The residual is computed from $\hat{\mathbf{x}}_2$, interpolated at full resolution:

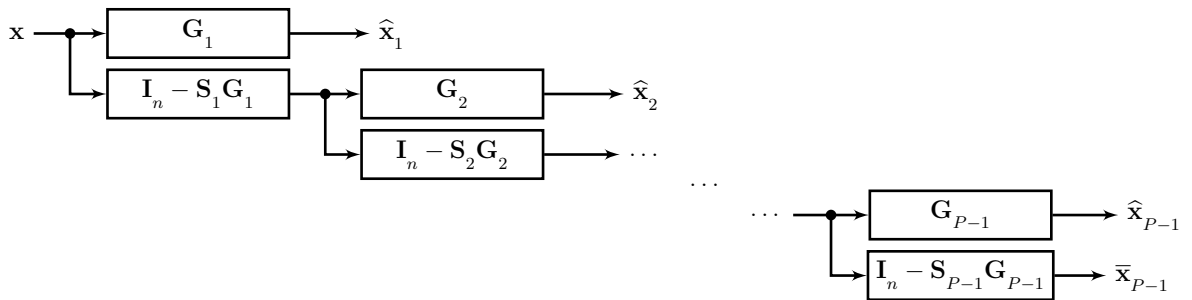
$$\bar{\mathbf{x}}_2 = \bar{\mathbf{x}}_1 - \mathbf{S}_2 \hat{\mathbf{x}}_2 \quad (15)$$

- The process continues until the low-pass filter $P - 1$:

$$\hat{\mathbf{x}}_{P-1} = \mathbf{G}_{P-1} \bar{\mathbf{x}}_{P-2} \quad (16)$$

$$\bar{\mathbf{x}}_{P-1} = \bar{\mathbf{x}}_{P-2} - \mathbf{S}_{P-1} \hat{\mathbf{x}}_{P-1} \quad (17)$$

This construction can be illustrated as follows:



From this illustration, it becomes obvious that the reconstruction of $\mathbf{x}$ from the successive components is given by:

$$\mathbf{x} = \sum_{p=1}^{P-1} \mathbf{S}_p \hat{\mathbf{x}}_p + \bar{\mathbf{x}}_{P-1} \quad (18)$$

Formally, the linear transform $\mathbf{F}$ can be written as:

$$\mathbf{F} = \begin{pmatrix} \mathbf{G}_1 \\ \mathbf{G}_2 (\mathbf{I}_n - \mathbf{S}_1 \mathbf{G}_1) \\ \vdots \\ \mathbf{G}_{P-1} \prod_{p=1}^{P-2} (\mathbf{I}_n - \mathbf{S}_p \mathbf{G}_p) \\ \prod_{p=1}^{P-1} (\mathbf{I}_n - \mathbf{S}_p \mathbf{G}_p) \end{pmatrix} \quad (19)$$

where:

- The sign $\prod$ should be understood as:

$$\prod_{p=1}^{P-1} (\mathbf{I}_n - \mathbf{S}_p \mathbf{G}_p) = (\mathbf{I}_n - \mathbf{S}_{P-1} \mathbf{G}_{P-1}) (\mathbf{I}_n - \mathbf{S}_{P-2} \mathbf{G}_{P-2}) \dots (\mathbf{I}_n - \mathbf{S}_1 \mathbf{G}_1) \quad (20)$$

- The total size of the transformed space is $\hat{n} = \sum_{p=1}^{P-1} n_p + n$.

The left-inverse of $\mathbf{F}$ is simply:

$$\mathbf{F}^\dagger = (\mathbf{S}_1 \mathbf{S}_2 \dots \mathbf{S}_{P-1} \mathbf{I}_n) \quad (21)$$

4 Scale-dependent localization

The scale-dependent localization (SDL) described in Buehner and Shlyueva (2015) can be seen as a special case of filter banks localization (FBL), where:

- All the components have the same grid size $\hat{n}_p = n$, so the size of the transformed space is $\hat{n} = Pn$.
- There is no interpolation: $\mathbf{S}_p = \mathbf{I}_n$

Many implementations are possible for the low-pass filter $\mathbf{G}_p$. In the original paper, it is defined in spectral space:

$$\mathbf{G}_p = \mathbf{T}^{-1} \mathbf{D}_p \mathbf{T} \quad (22)$$

where:

- $\mathbf{T}$ is a spectral transform.
- $\mathbf{D}_p$ is the diagonal analytical filter defining waveband p .

5 Interpolation-based filter banks localization

The interpolation-based filter banks localization (IFBL) is a special case of the filter banks localization (FBL) where the low-pass filter $\mathbf{G}_p$ for component p is given by:

$$\mathbf{G}_p = \mathbf{N}_p \mathbf{S}_p^T \quad (23)$$

where the i^{th} coefficient of the diagonal normalization matrix $\mathbf{N}_p$ is equal to the inverse of the i^{th} coefficient of the vector $\mathbf{S}_p^T \mathbf{1}_n$.

6 Methods comparison

To illustrate the similarities and differences between the SDL and IFBL methods, we start from a random field $\mathbf{x}$ with different scales depending on the location in Figure 1. In Figure 2, the filtered field $\hat{\mathbf{x}}_1$ at low resolution captures the largest scales of the signal. These scales are removed in the corresponding residual $\bar{\mathbf{x}}_1$, in Figure 3. In Figure 4, the filtered field $\hat{\mathbf{x}}_2$ at intermediate resolution captures the intermediate scales of the signal. Only the smallest scale are left in the final residual $\bar{\mathbf{x}}_2$, in Figure 5.

Here, the SDL uses the `scipy.ndimage.gaussian_filter` function of Python for $\mathbf{F}_p$. The IFBL uses a standard bilinear interpolation. The filtering scales of the SDL and the low-resolution grids of the IFBL have been briefly tuned to yield visually similar results.

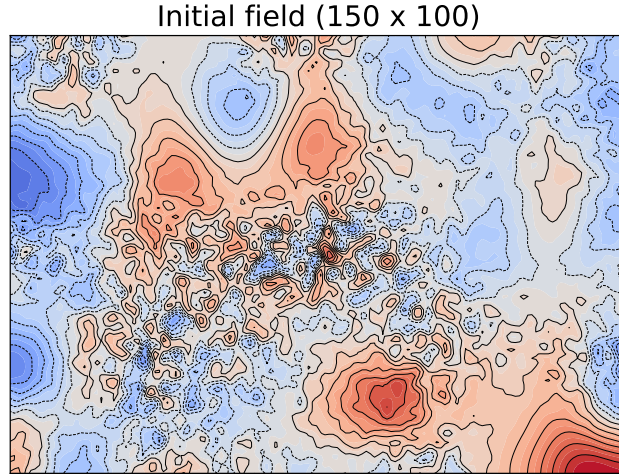


Figure 1: Random field with different scales depending on the location.

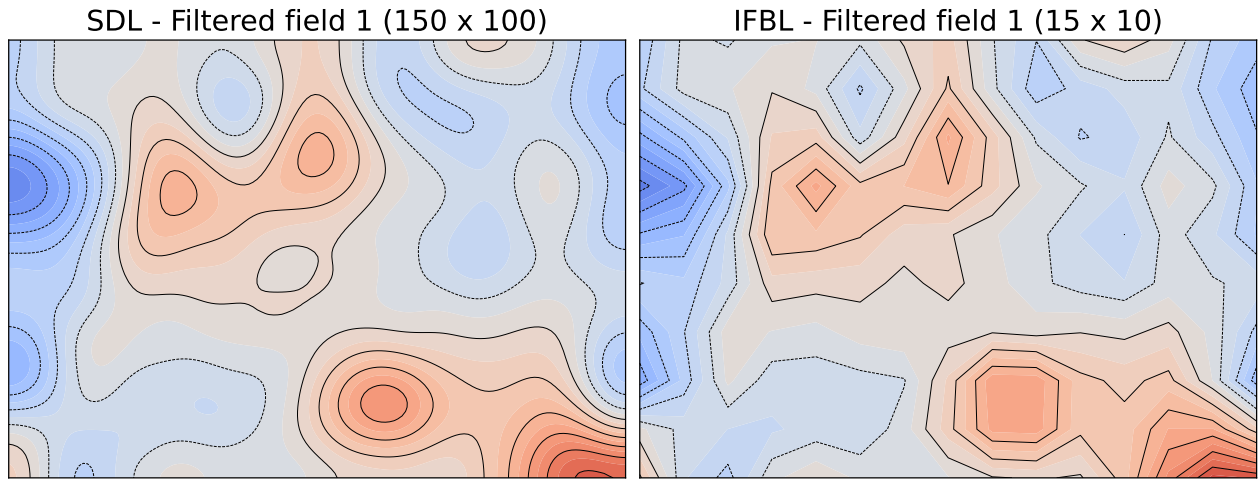


Figure 2: Filtered field at low-resolution 1 with SDL (left) and IFBL (right) and corresponding residual (right).

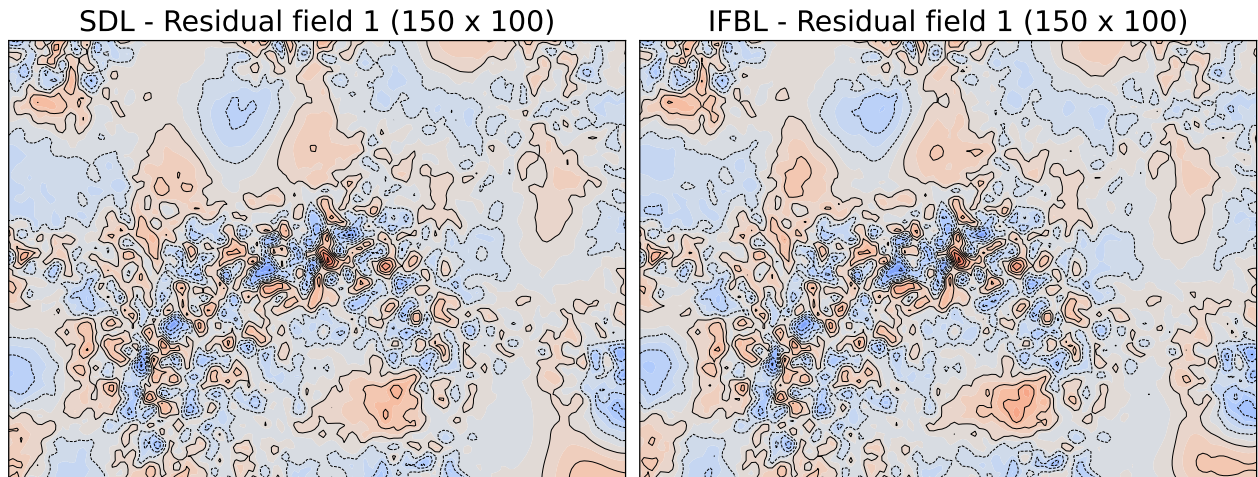


Figure 3: Residual field at low-resolution 1 with SDL (left) and IFBL (right).

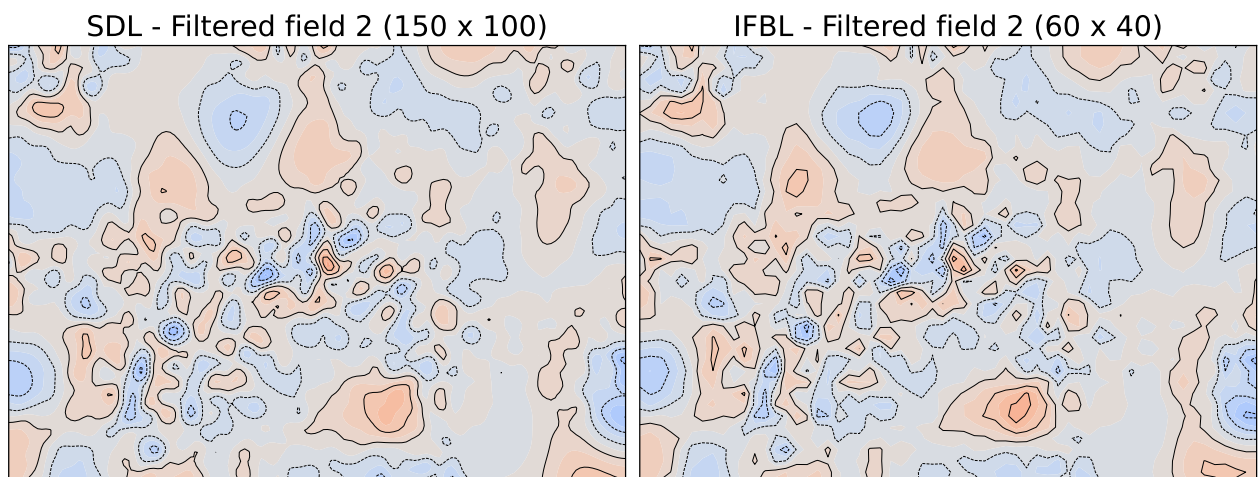


Figure 4: Filtered field at low-resolution 2 with SDL (left) and IFBL (right) and corresponding residual (right).

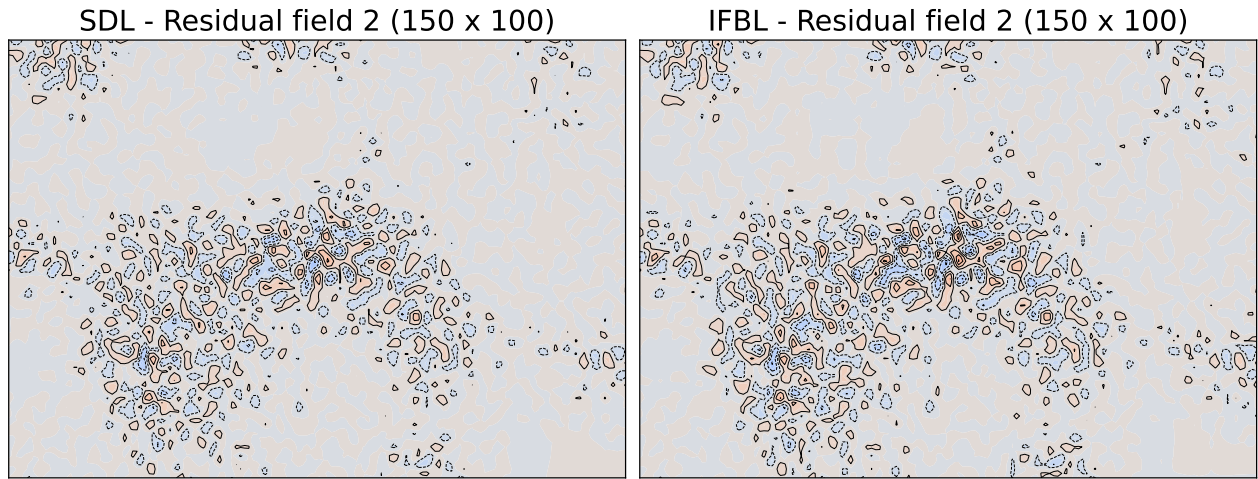


Figure 5: Residual field at low-resolution 2 with SDL (left) and IFBL (right).

The interpolation-based filter banks localization (IFBL) leads to a smaller size of the transformed space $\hat{n}$ than the scale-dependent localization (SDL), reducing the required memory for the same number of components. Also, the choice of the grids for the low-resolution components is completely flexible. Once the interpolation from low resolution to full resolution and its adjoint are available, the construction of the low-pass filter is straightforward. However, the IFBL requires that the localization matrix is applied in the geometry of each component, which is more complicated but also cheaper because the localization is applied at lower resolution.

References

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- Kleist DT, Ide K. 2014. An OSSE-Based evaluation of hybrid variational-ensemble data assimilation for the NCEP GFS. Part I: System description and 3D-hybrid results. *Monthly Weather Review* **143**(2): 433–451.